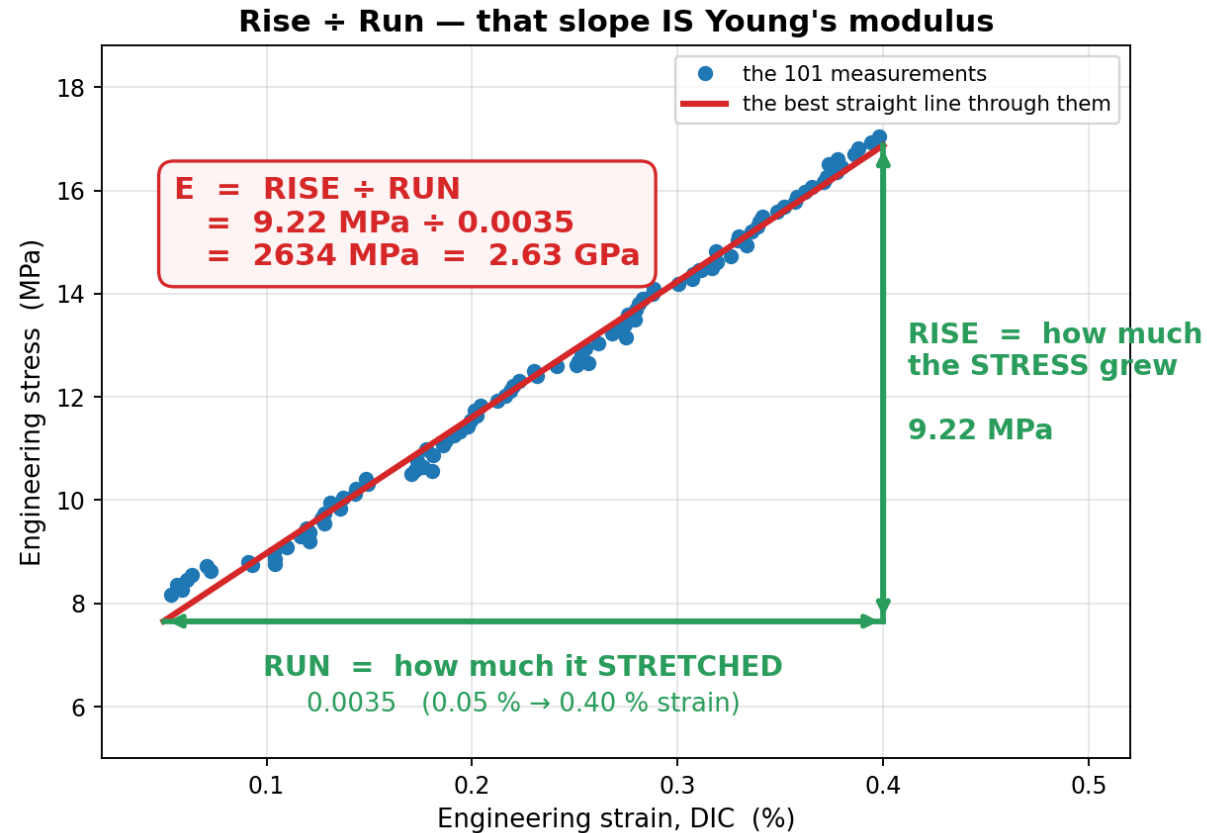


Young's modulus E — how stiff is the material?

Pull the specimen. It stretches. E is how steeply the pull has to rise to keep it stretching.



STRESS = how hard you are pulling ÷ how thick the specimen is.
Units MPa.

STRAIN = how much the gauge stretched ÷ how long it was.
Just a fraction — no units.

E = how much the STRESS went up, divided by how much it STRETCHED to get there.

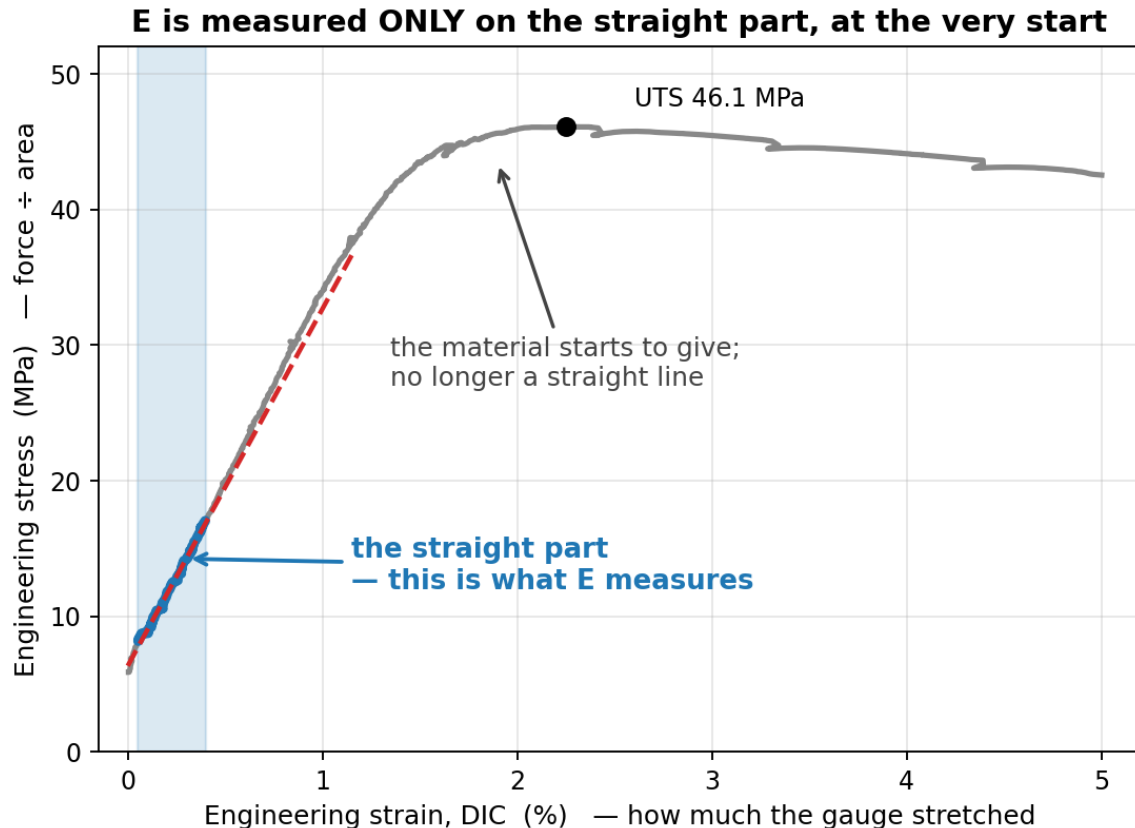
$$\begin{aligned} \text{RISE} &= 16.87 - 7.65 = 9.22 \text{ MPa} \\ \text{RUN} &= 0.0040 - 0.0005 = 0.0035 \\ E &= 9.22 / 0.0035 = 2634 \text{ MPa} \end{aligned}$$

2634 MPa = 2.63 GPa. A stiffer material gives a steeper line and a bigger E ; a floppier one gives a shallower line.

Steepness of the straight bit = stiffness. Nothing else on the curve is used for E .

The formula the code actually uses

Two points would do — but 101 were measured, so all 101 are used. That is all "least squares" means.



Best-fit slope through all the points:

$$E = \frac{n \cdot \sum(\epsilon \cdot \sigma) - \sum \epsilon \cdot \sum \sigma}{n \cdot \sum(\epsilon \cdot \epsilon) - (\sum \epsilon)^2}$$

Add up four columns over the 101 points:

n	=	101
$\sum \epsilon$	=	0.237696
$\sum \sigma$	=	1266.1435 MPa
$\sum(\epsilon \cdot \epsilon)$	=	0.000657593
$\sum(\epsilon \cdot \sigma)$	=	3.238406 MPa

Put the numbers in:

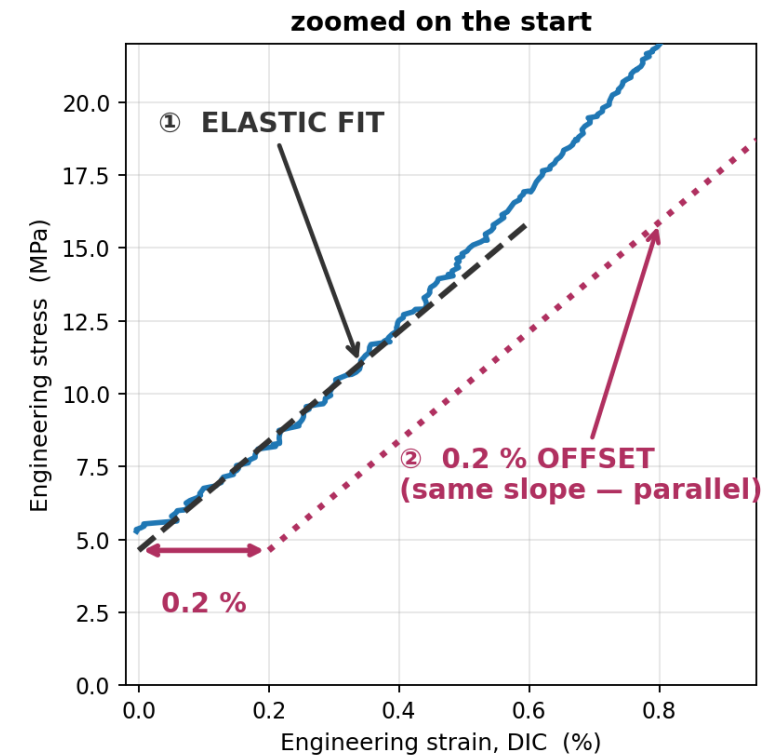
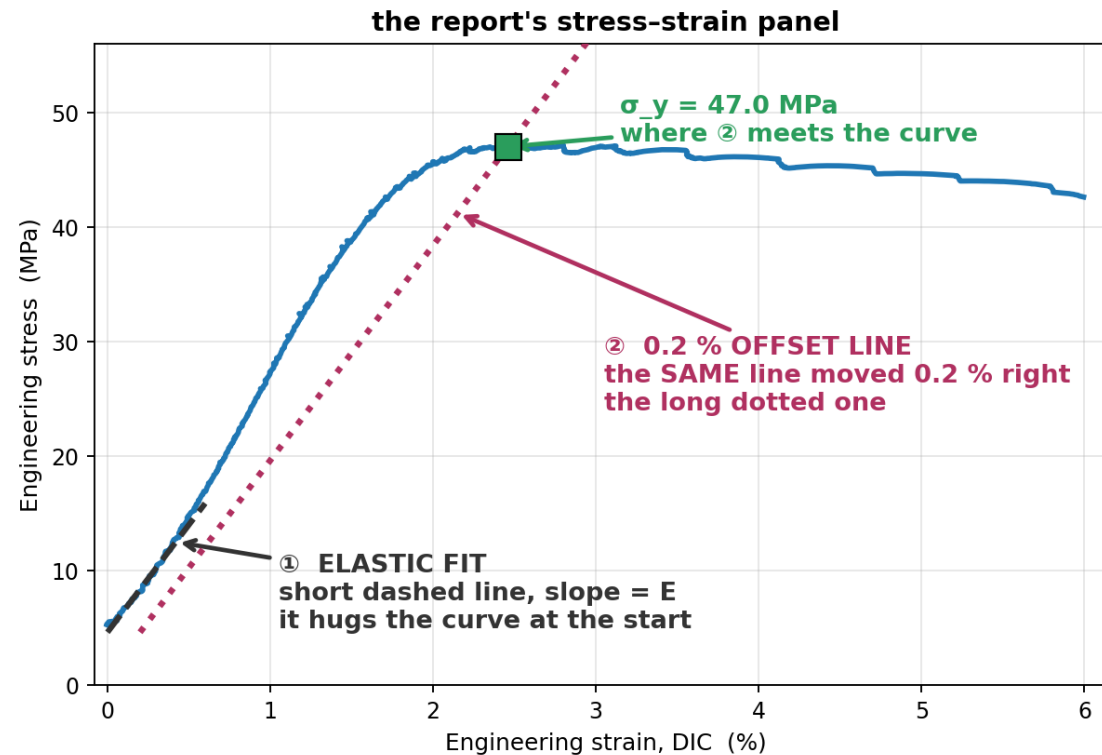
$$\begin{aligned} \text{top} &= 101 \cdot 3.2384 - 0.2377 \cdot 1266.14 = 26.1218 \\ \text{bottom} &= 101 \cdot 0.0006576 - 0.2377^2 = 0.0099175 \\ E &= 26.1218 / 0.0099175 = 2634 \text{ MPa} = 2.63 \text{ GPa} \end{aligned}$$

$$\sigma = 2634 \times \epsilon + 6.34 \quad R^2 = 0.9936 \quad \text{the 6.34 MPa offset is the preload the curve starts from, not an error}$$

The two grey lines on the report's stress–strain graph

Both have the SAME slope — the one E you measured. Neither of them is used to compute E .

① and ② are the SAME slope (= E). ② is just shifted 0.2 % to the right.



1 Fit E first — least squares over the points from 0.05 % to 0.40 % strain.

2 Draw ① with that slope: the short dashed line that hugs the start of the curve.

3 Copy it and slide it 0.2 % to the right — that copy is ②, the long dotted line.

4 Where ② meets the curve is σ_y , the 0.2 % offset yield stress.

The line is not an extrapolation FROM the yield point — the yield point is defined BY the line. $S16 \cdot E$ 1.88 GPa · σ_y 47.0 MPa